Machines

MACHINES MAKE WORK EASIER TO DO.

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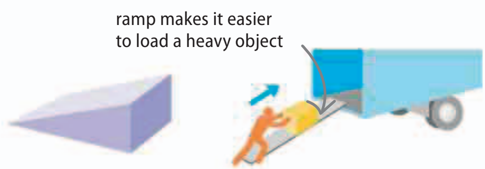
A simple machine is a device that increases the size, or changes the direction, of a force. Machines can use this energy to lift, cut, or move masses.

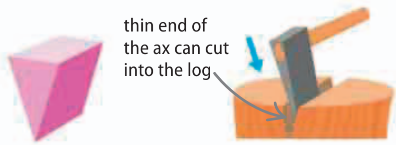
Simple machines

Even the most complex devices can be broken down into half a dozen simple machines working together. All of these machines have been in use since ancient times, and at first glance some may not seem to be machines at all. However, the way they all multiply the force applied to them, or multiply the distance over which that force acts, makes them machines.

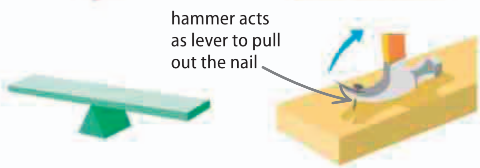
▷ **Ramp**
A heavy load can be pushed up a ramp in a continuous motion that requires a smaller force than lifting it straight up.



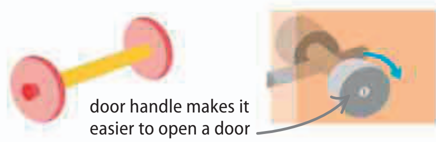
▷ **Wedge**
Any force applied to the thick end of a wedge is focused into the thin end, applying enough pressure to cut into materials.



▷ **Lever**
When a force is applied, the lever moves around a turning point (fulcrum), creating an opposite force at a different point along the lever.

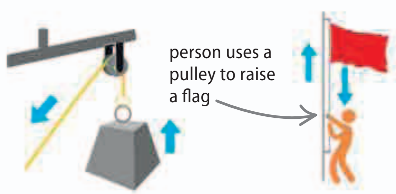


▷ **Wheel and axle**
A wheel moves around an axle in the same way as a lever around a fulcrum, multiplying the distance over which a force acts.



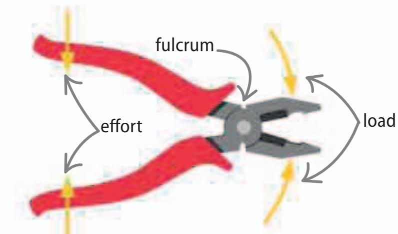
▷ **Screw**
A screw wraps around an axle, so the load moves up the screw as it turns. The tip of a screw is often pointed, so it also acts as a wedge.

▷ **Pulley**
A pulley is a rope looped around one or more wheels. The pulley's wheel changes the direction of the force on the rope—so pulling down makes the weight go up.

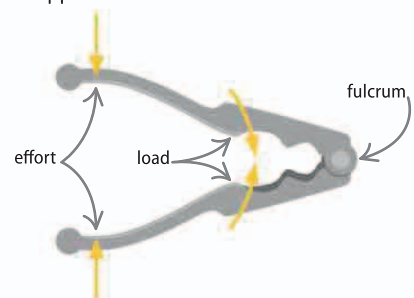


Levers

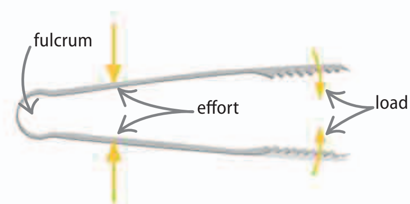
Levers magnify a small force into a large force. They work by moving a load around a turning point. There are three types of levers; the difference between them depends on where the effort, load, and fulcrum are positioned.



△ **First-class lever**
When you use pliers, the effort is applied on the opposite side of the fulcrum to the load.



△ **Second-class lever**
When you use a nutcracker, the load is positioned between the effort and the fulcrum.



△ **Third-class lever**
When you use tongs, the effort is applied between the load and the fulcrum.